Causal Probabilistic Transformer: experimental report, 2026-08-09 – 2026-08-11

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2026-08-11

Abstract

The causal PT decoder was implemented from scratch, validated against the worked example of the technical note digit for digit, and trained on PTB. Best replicated result: validation perplexity 250.69 ± 2.32, test 229.31 ± 2.62 (three seeds), against a unigram baseline of 688.82, a GPT baseline of 115.43 and a Looped-Transformer baseline of 126.46 on the identical pipeline and scored token set — at 16,448 non-embedding parameters against the GPT's 1,237,440. This document is the full record: every experiment, every number, the failed diagnoses (kept and marked SUPERSEDED), the corrections log, and the reproduction map. Every number is sourced from a run JSON, a slurm log, or a checkpoint inspection. Single-seed results are labelled as such throughout. Repository: `https://github.com/b3ly4ck/probabilistic-transformer-experiments` (commits `ed8dda7`... `d437a76`).

1 Executive summary

model	val ppl	test ppl	emb / non-emb params	seeds
Causal PT, best (d=32, h=2, α_Z=0.25, frozen b)	250.69 ± 2.32	229.31 ± 2.62	330,000 / 16,448	3
Causal PT, best 2×2 cell (floor 0.5 + G_t)	247.66 ± 5.36	225.52 ± 5.46	330,000 / 18,496	3
Looped Transformer (1 shared block × 4)	126.46	117.93	1,610,240 / 309,600	1
GPT (nanoGPT, 4 layers)	115.43	107.16	1,610,240 / 1,237,440	1
unigram baseline	688.82	—	—	exact

The PT is 2.17× worse than the GPT on validation at 75× fewer non-embedding parameters. It cannot currently be given a comparable budget: without damping it collapses above $d = 24$ labels and above $h = 2$ channels, and with damping the d-ladder is non-monotone (open). The matched-budget comparison the research plan requires therefore does not exist yet; the honest object is the trade curve of Fig. 6, which is what §22.3 of the technical note anticipated.

Three mechanisms were established.

(1) The message/unary balance decides learning, and the transition is visible within a run. The scalar $\|G\|/\|S_w\|$ (head message against word unary in the content stream) separates every run that learns from every run that lands on the unigram, across ~30 PTB runs without exception. In run 940844 the model sits at the unigram for 2000 steps and validation perplexity drops $703.6 \to 539.0$ at the very evaluation where $\|G\|/\|S_w\|$ first enters the healthy range; in run 940848 an excursion out of the range at steps 1500–2000 coincides with perplexity *reverting* $516 \to 694$, then resuming its fall when the scalar re-enters. The dependence is two-sided (Figs. 1, 2).

(2) A step size below 1 on the label update (Wu & Tu App. B.1) breaks the runaway loop and opens $d = 32$. The loop $\bar{q} \to B \to G \to \bar{q}$ explodes at $d = 32$ ($\|G\|/\|S_w\|$ $2.88 \to 20.57$ between steps 500 and 1000); $\alpha_Z = 0.25$ damps it and takes validation perplexity from 315.46 (the d=16 ceiling) to 250.69 ± 2.32. This is the largest single gain of the session and is replicated over three seeds. $\alpha_Z = 0.5$ is not enough (531.10, single seed).

(3) The B.3.3 global head dies by a symmetry trap and revives only when treated unlike the other factors. Under uniform initialisation and a shared L2, the rows of B' collapse to one row replicated m times (row spread $0.0197 \to 0.0037$ while within-row structure stays 0.90); the gradient argument is two lines (§5.5). With asymmetric initialisation and no L2 on B' the head carries half the total message and improves an undamped $d = 24$ by 7.4 %. On the damped record it *costs* 7.7 % and doubles the seed variance. Both readouts defeat it for different reasons: under the exact readout its contribution is a constant by construction (measured to 10^{-12}); under MFVI it is degenerate by optimisation unless specially treated.

Interaction claim, withdrawn. On single seeds, G_t + a raised cosine floor beat the record (241.89 vs 248.29) while each intervention alone was worse — an apparent interaction. Replication over three seeds gives 247.66 ± 5.36 against 250.69 ± 2.32: a 1.2 % difference against overlapping ranges. The single-seed record of 241.89 is withdrawn; the defensible number is the damped $d = 32$ figure above.

2 Setup

Data. PTB word level, Mikolov preprocessing, one sentence per line, `<eos>` appended. Vocabulary from train: 10,000 types. Tokens: train 929,589, valid 73,760, test 82,430. Fixed-length blocks of 64 tokens from the joined stream, identical for every model.

Unigram floor. ML unigram from train counts (no smoothing; the vocabulary is built from train), scored on the identical token set (deterministic non-overlapping validation blocks, first slot dropped): val **688.82** (687.45 with slot 0). The pre-registered gate was half of it, 344.41.

Evaluation protocol. PT's logits[:, t] predicts w_t from $w_{<t}$ (slot 0 from ROOT alone). The GPT wrapper runs the backbone on idx[:, : −1] and shifts its output right by one slot, so both models answer $p(w_t \mid w_{<t})$; slot 0 of the GPT is zeros and its loss asserts it is never scored. `ignore_first`=1 everywhere makes the scored token sets identical (asserted by a test). Evaluation is deterministic (fixed blocks, whole split); train perplexity is measured on a fixed 20-batch slice the same way. The context probe (prefix ablation) compares the last-slot predictive distribution under the true prefix, a shuffled prefix, and a constant prefix (KL, max $|\Delta \text{logit}|$, argmax agreement).

Shared loop. One implementation for all models; model-specific behaviour only via `loss`, `arc_regulariser`, `num_parameters`. AdamW, $\beta = (0.9, 0.999)$, weight decay $1.4{\cdot}10^{-6}$, gradient clip 1.0, warmup 100 + cosine to floor × lr (floor 0.1 unless stated), L2 on the arc scores $5{\cdot}10^{-4}$ (both from Wu & Tu Table 2, PTB). Dropout is not implemented for PT (placement unspecified by the source; any placement inside the updates crosses the §22.2 "no maps" tripwire); the GPT ran with dropout 0.

PT fields. d = label-set size of Z_t; h = head channels; rank = Kruskal rank of $T^{(c)} = U^{(c)} V^{(c)\top}$; γ = RPE clip (causal half, γ+1 buckets); T = content-stream MFVI iterations (layer-parallel); τ = predictive inner rounds of the MFVI readout; $\lambda_Z = 1$, $\lambda_H = 1/d$, $\lambda_W = 1$, $\lambda_G = 1$ (unpinned by the source); α_Z = App. B.1 step size (1 = original update, asserted bitwise); init_std = 0.02 (nanoGPT convention; specified by neither paper); freeze-b = clamp the word unary to corpus log-frequencies, never updated; m (`n_global`) = size of the B.3.3 single-split global head.

Hardware. Slurm partition `critical`; NVIDIA TITAN RTX (24 GB) or A40 (46 GB) per job as listed in §8; torch 2.0.1+cu117. Nodes `ai_gpu32/33` do not mount `/public` and killed six jobs silently before being identified (§7).

3 Baselines

Both ran in job 940799 (A40), 6000 steps, identical loop, blocks, token set, and unigram reference. GPT is nanoGPT off the shelf (n_{embd}=160, 4 layers, 4 heads, tied embeddings). Looped is the same with one shared block applied four times: PT's weight sharing without PT's structure.

GPT at 115.43 *exonerates the pipeline* (the predecessor implementation's own GPT reached 131 on its own pipeline). Looped at 126.46 *exonerates weight sharing*: it has PT's parameter sharing, none of its structure, 3.4× fewer non-embedding parameters than the PT control, and it works. The PT ablation row is the failure in one line: the trained control's output does not depend on the prefix at all.

4 The failure and its diagnosis

4.1 The sweep: every early configuration lands on the unigram

Base: d=256, h=8, rank=64, γ=3, T=3, τ=2, MFVI, lr 10^{-3}, 6000 steps (exact-readout run: 2000), seed 0. Unigram 688.82.

Table 1: Headline configurations, full dumps. Shared: batch 16, block 64, warmup 100, weight decay $1.4 \cdot 10^{-6}$, clip 1.0, $\ell_2^{\text{arc}} = 5 \cdot 10^{-4}$ (PT), init_std 0.02, root_init_std = init_std, $\lambda_Z = \lambda_W = \lambda_G = 1$, $\lambda_H = 1/d$, T=3, τ=2, MFVI readout, seed 0 unless stated.

field	pilot	GPT	Looped	3a d=16	ladder d=16	**record** d=32	G_t rev. d=24	2×2 best
job(s)	940422	940799	940799	940844	940858	940914/932/934	940918	940921/931/933
d/h/rank/γ	256/8/64/3	n_{embd}160, L4, H4		16/2/16/3	16/2/16/3	32/2/32/3	24/2/24/3	32/2/32/3
α_Z	1.0	—	—	1.0	1.0	**0.25**	1.0	0.25
m; B' init; B' in L2	0	—	—	0	0	0	64; 0.5; no	64; 0.5; no
freeze b	no	—	—	no	**yes**	**yes**	yes	yes
lr / floor / steps	1e-3/0.1/6k	1e-3/0.1/6k	same	2e-2/0.1/6k	2e-2/0.1/15k	2e-2/0.1/15k	2e-2/0.1/15k	2e-2/**0.5**/15k
seeds	0	0	0	0	0	0,1,2	0	0,1,2
emb / non-emb	2.57M/1.05M	1.61M/1.24M	1.61M/0.31M	170k/4,128	170k/4,128	330k/16,448	250k/10,800	330k/18,496
wall-clock	308 s	123 s	116 s	203 s	483 s	656/536/538 s	546 s	626/594/687 s
hardware	TITAN	A40	A40	A40	A40	TITAN	TITAN	TITAN
val / test	695.70/655.09	115.43/107.16	126.46/117.93	473.28/433.06	315.46/285.33	250.69 ± 2.3/229.31 ± 2.6	297.71/272.82	247.66 ± 5.4/225.52 ± 5.5

model	val	test	emb / non-emb	ablation (shuffled): KL / max$|\Delta|$ / argmax same
GPT	115.43	107.16	1,610,240 / 1,237,440	3.9461 / 14.04 / 0.117
Looped	126.46	117.93	1,610,240 / 309,600	3.6469 / 14.91 / 0.125
PT control (d=256, h=8, lr 1e-3)	695.84	655.25	2,570,000 / 1,050,624	0.0000 / 0.0000 / 1.000

job	change from base	val	test	wall	HW
940422	none (pilot)	695.70	655.09	308 s	TITAN
940423	exact readout, 2000 steps	1555.04	1508.89	794 s	TITAN
940436	$T = 1$	695.33	654.60	181 s	TITAN
940435	$\ell_2^{\text{arc}} = 5.0$	691.82	649.88	307 s	TITAN
940438	$\lambda_Z = 4$	690.02	646.51	230 s	A40
940483	$\lambda_Z = 16$	696.24	655.82	287 s	TITAN
940484	$\lambda_Z = 32$	697.76	657.63	261 s	TITAN
940485	$\lambda_Z = 64$	699.29	659.42	228 s	A40
940444	word unary b removed	985.02	945.01	291 s	TITAN
940445	control for 940444	695.73	655.10	264 s	TITAN
940489	init_std = 0.2	695.53	654.86	298 s	TITAN
940490	init_std = 0.5	702.57	660.80	263 s	TITAN
940502	$\tau = 1$ (init 0.5)	700.45	659.64	258 s	TITAN
940503	$\tau = 4$ (init 0.5)	696.00	653.52	347 s	TITAN
940504	$\tau = 8$ (init 0.5)	709.45	668.56	434 s	A40

Fifteen configurations spanning 45× in λ_Z, 10^4× in the L2 coefficient, 25× in init_std, $T \in \{1, 3\}$, $\tau \in \{1, 2, 4, 8\}$, b on/off, both readouts: every MFVI run within 1–3 % of the unigram. Perplexity falls ($5692 \to 695$ within run 940422) but falls *to* the unigram.

Diagnoses raised and killed, in order:

superseded — "b is a shortcut to the unigram." Removing b made the model worse (985 vs 696). (Dropping b never made the unigram unreachable: with μ constant, $\text{logits}(w) = \text{LSE}_a(S_{w,a} + \log \mu(a))$ is still a function of w alone.)

superseded — "the label softmax saturates." ($\|G\| = 45$ vs $\|\bar{s}\| = 0.74$, label entropy 0.0009 nats.) Registered prediction: raising λ_Z should recover the entropy *and* make the ablation KL non-zero. Measured: entropy recovered fully ($0.044 \to 4.95 \to 5.26 \to 5.48$ nats at $\lambda_Z = 1/16/32/64$), perplexity unmoved (695.73/696.24/697.76/699.29), ablation KL exactly 0 throughout. The position-variance of $\bar{q}$ *worsened* ($3 \cdot 10^{-5} \to 4 \cdot 10^{-11}$).

superseded — "the initialisation is context-blind." True as a measurement (untrained ablation KL $9.5 \cdot 10^{-11}$ at init 0.02 vs $6.9 \cdot 10^{-2}$ at 0.5) but not binding: training from the context-using initialisation still landed on the unigram (702.57).

The identity that closes the case. In every failing run $\bar{q}$ is nearly identical across positions (std 0.0008–0.0013 vs uniform $1/256 = 0.0039$). If $\bar{q}_j$ is the same for all j, then $B^{(c)}_{j,a}$ is the same for all j, and

$$\log \mu_t(a) \;=\; \textstyle\sum_c \text{LSE}_{j \in D_t} B^{(c)}_{j,a} \;=\; \sum_c \left[B^{(c)}_a + \log |D_t|\right],$$

where $\log |D_t|$ is constant in a and is removed exactly by the readout's normalisation. The logits then cannot depend on prefix content — only on t — and the model is left with $p(w \mid \text{position in block})$, whose average is the unigram. Measured directly: logits vary 1700× more across slots than across sequences (std $1.5 \cdot 10^{-1}$ vs $8.9 \cdot 10^{-5}$).

4.2 Stage-by-stage content fraction: no single guilty stage

Content fraction = std over sequences at a fixed slot ÷ overall std. Control: an *untrained* model at init 0.5, whose readout demonstrably works (ablation KL $6.9 \cdot 10^{-2}$).

The architecture transmits word identity end to end (control column). Every trained model ends at 10^{-5}–10^{-9}, but the content dies at *different* stages per configuration; no single operation is broken.

stage	untrained 0.5	trained init 0.5	trained λ_Z=16	trained baseline
$\bar{q}$ (content stream)	3.10e-01	3.61e-02	5.18e-02	9.10e-04
B (contracted arcs)	7.68e-01	2.41e-01	—	2.74e-03
G round 1	6.28e-01	4.75e-01	**6.00e-05**	3.08e-03
Q_Z round 1	3.22e-01	**5.87e-04**	6.52e-05	7.15e-09
logits	**7.35e-01**	9.33e-05	2.98e-09	1.33e-08

4.3 Scale bisection with GPT control — superseded as a scale story

Order-1 Markov chain, 5 successors per state; oracle = visit-weighted mean conditional entropy; add-1-smoothed unigram; 3000 steps, lr 10^{-3}; gap closed = (unigram − best)/(unigram − oracle). Job 940799.

cell	oracle	unigram	GPT gap closed	PT gap closed
V=11, d=256	3.71	7.84	1.149	0.000
V=100, d=256	3.65	81.19	1.001	0.000
V=1000, d=256	3.61	795.42	1.000	0.000
V=10000, d=256	3.61	7945.95	1.000	0.018
V=1000, d=16	3.61	795.42	0.984	0.062
V=1000, d=64	3.61	795.42	1.000	0.044

(GPT > 1 is possible: the oracle is an estimate, not a bound.) "The failure is scale-dependent" is SUPERSEDED: at lr 10^{-3} PT extracts nothing at any vocabulary and any width — including $V = 11$, where the next token is fixed by one preceding symbol out of eleven — while GPT reaches the oracle everywhere. What replaced it: the failure is *joint* in (d, lr).

4.4 The lr probe and the $d \times \mathrm{lr}$ grid

lr probe (job 940809; chain oracle 3.84 / unigram 6.52, pre-v0.8.2 chain):

lr	0.001	0.005	0.02	0.05	0.1
PT d=16, gap closed	0.069	0.546	**1.011**	0.000	0.009
PT d=256	0.000	0.000	0.000	0.000	0.115
GPT d=256 (control)	1.350	1.350	1.323	0.849	0.321

At d=16, lr=0.02 the PT reaches the oracle (3.809 vs 3.84). PT needs a rate 20× above the Table 2 value — at which the GPT already degrades — and the small width at that rate. All fifteen sweep configurations sat outside this region on both axes at once. The $d \times \mathrm{lr}$ grid (job 940814, different chain instance, oracle 1.96 / unigram 3.09; fractions comparable across probes, raw perplexities not) is Fig. 3; best cell d=16, lr=0.005 replicated over seeds 0/1/2: gap closed 0.936 / 0.886 / 0.780. Caveat: this grid varied h together with d ($h = 8$ if $d \geq 64$ else 2), deconfounded next.

4.5 The $d \times h$ deconfound

Full cross grid, lr 0.005 fixed, one chain (oracle 3.577 / unigram 7.105, post-v0.8.2), seed 0, 3000 steps per cell, job 940827. Fig. 4.

Channels degrade the model monotonically at every width; width degrades it independently of channels (at h=2: 0.405 → 0.724 → 0.023 → 0.000). Neither may be attributed to the other. The bound behind the channel effect: $G_i(a) = \sum_c \sum_{j \in D_i} Q_c(j)\, B^{(c)}_{j,a}$ with Q_c a distribution and each $B^{(c)}_{j,\cdot}$ an expectation of a row of $T^{(c)}$ under $\bar{q}_j$ (ROOT row = $r^{(c)}$), hence $|G_i(a)| \leq h \cdot \max(\max|T|, \max|r|)$ — linear in h, independent of d.

5 The fixes, in causal order

All PTB, $h = 2$, MFVI readout, lr 0.02, warmup 100, seed 0 unless stated.

d	h	val	train	gap closed	$\|G\|/\|S_w\|$	$H(q)$	ablation KL
16	2	5.675	7.006	0.405	8.56	0.122	6.93e-02
16	4	6.715	6.671	0.110	18.81	0.089	1.45e-01
16	8	6.502	6.462	0.171	15.52	0.118	1.59e-01
32	2	4.551	4.485	**0.724**	6.78	1.796	9.31e-01
32	4	5.003	4.921	0.596	8.21	0.950	6.80e-01
32	8	7.104	7.047	0.000	36.68	0.083	0.000
64	2	7.023	6.974	0.023	22.77	1.079	1.69e-04
64	4	7.103	7.044	0.001	43.48	0.105	0.000
64	8	7.106	7.051	0.000	47.51	0.047	0.000
128	2	7.106	7.050	0.000	12.21	0.156	0.000
128	4	7.106	7.051	0.000	28.39	0.118	0.000
128	8	7.106	7.051	0.000	25.78	0.036	0.000

5.1 Freeze b at the corpus log-unigram

b_w represents the unigram exactly, so the fastest descent direction is to fit the marginal through b while the context path idles (run 940844 sits at the unigram for 2000 steps). Prepaying the marginal (set $b = \log \hat{p}_{\text{unigram}}$, never update; the factor stays in the graph, observed rather than learned) removes the race.

run	job	steps	b	val	train	test	$\|G\|/\|S_w\|$	abl. KL
3a	940844	6000	learned	473.28	488.85	433.06	3.41	0.913
3c	940848	6000	frozen	430.04	450.35	393.09	3.96	0.799

$-9.1\,\%$ val at identical budget (single seeds); the phase transition moves from step 2500 to step 1000. The excursion trace (Fig. 2) is the two-sided evidence: $\|G\|/\|S_w\|$ rises to 8.64/11.70 at steps 1500–2000 and val *reverts* 516 $\to$ 694, then falls monotonically to 430 once the scalar re-enters. Freezing b also lifts the label entropy from 0.030 to 0.870 (of 2.77): the free b was driving $\bar{q}$ into degeneracy, not merely spending steps.

5.2 Budget, with the schedule caveat

15000 steps with b learned: 336.89 / 338.90 / 308.04 (val/train/test), job 940850; with b frozen, d=16: 315.46 / 314.09 / 285.33, job 940858. The long run ends flat (last evals 339.19/338.84/336.89/337.21) where 6000 steps cut mid-descent. **Caveat, unresolved:** the cosine schedule is defined over max steps, so the long run is not "the same run, longer" — at step 6000 its lr is still high while the short run has decayed to 0.1×. Budget and schedule are not separated by these runs. Composition: 473.28 $\to$ 430.04 (freeze) $\to$ 336.89 (budget) $\to$ 315.46 (both); all single seeds. First ladder at this configuration (940858): d=24: 321.43 / 325.84 / 294.89 ($\|G\|/\|S_w\|$ 2.26, KL 2.003); d=32: **final 641.5** — the stored 523.08 is the trace minimum, reached at step 500 before the message exploded (§7).

5.3 Damping opens $d = 32$

The runaway loop: larger T $\to$ sharper attention $\to$ the message becomes a full-magnitude B row $\to$ $\bar{q}$ sharpens $\to$ T grows. App. B.1 step size $Q^{(t)} = \alpha_Z Q^* + (1 - \alpha_Z)Q^{(t-1)}$ on the label update damps exactly this loop; $\alpha_Z = 1$ reproduces the original update bitwise (asserted).

run	job	HW	val (final)	train	test	$\|G\|/\|S_w\|$	abl. KL	non-emb
d=32, α_Z=1	940858	A40	641.5	664.95	582.92	5.33	0.281	16,448
d=32, α_Z=0.5	940913	TITAN	531.10	554.91	483.35	4.90	0.701	16,448
d=32, α_Z=0.25	940914	TITAN	**248.29**	225.43	226.62	1.31	2.851	16,448
d=48, α_Z=0.25	940915	TITAN	344.31	354.42	315.90	2.89	1.668	36,960
d=64, α_Z=0.25	940916	TITAN	271.41	253.69	249.64	2.56	2.694	65,664

Record replicated over seeds 1/2 (940932/940934): 250.86 / 252.93 $\Rightarrow$ 250.69 $\pm$ 2.32 val, 229.31 $\pm$ 2.62 test. Ablation KL rises 2.182 $\to$ 2.851 (more context used, not merely a better score); train $<$ val for the first time (225.43 vs 248.29) — the model crosses into overfitting, making regularisation meaningful for the first time. The threshold sits between 0.5 and 0.25: at 0.5 the damping stalls learning as well as the explosion. The ladder above 32 at fixed α_Z is *non-monotone* (48 worst; the only damped run with train $>$ val): read as "what $\alpha_Z = 0.25$ does across widths", not as a ceiling — the ceiling under damping is open.

5.4 The cosine floor and the 2×2

All cells d=32, α_Z=0.25, frozen b, 15000 steps (Fig. 5); val/test, seed 0:

	no G_t	with G_t (revived form)
floor 0.1	248.29 / 226.62 (940914)	267.51 / 245.69 (940920)
floor 0.5	251.80 / 230.80 (940927)	241.89 / 219.95 (940921)

On seed 0 each intervention alone is worse (G_t +7.7 %, floor +1.4 %) and the pair is better (−2.6 %) — an apparent interaction. Replication (seeds 1/2): best cell 248.61 / 252.48 ⇒ 247.66 ± 5.36; record cell ⇒ 250.69 ± 2.32. The 1.2 % mean difference sits inside overlapping spreads and the G_t cell has twice the seed variance. The interaction claim and the 241.89 record are **withdrawn**; what survives at replication strength is only that G_t at floor 0.1 hurts (+7.7 %).

5.5 The global head G_t, end to end

Form verification (before any run). B.3.3 single-split per Wu & Tu Eq. 46: one $B' \in \mathbb{R}^{m \times d}$, no channel axis; the message $\sigma(qB'^{\top}/\lambda_G)\,B'$ (the GFU operator, Eq. 62) is added once per round — asserted behaviourally (identical message under h=2 and h=8 with equal B'). B' was added to the L2 alongside T and r (the root-column lesson: an unpenalised carrier absorbs the message).

Exact-readout exclusion. Under the exact readout the head's direct contribution is $\log \mu_t(a)$ += $\mathrm{LSE}_k B'_{k,a}$ — position- and prefix-independent, measured constant to 10^{-12}; a run in that mode measures a label prior. All G_t runs are MFVI (enforced by the config).

First death (uniform treatment). Jobs 940872 (d=16) and 940879 (d=24), m=64, B' init 0.02, B' in the L2:

config	val	train	test	$H(Q_g)/\max$	glob msg	arc msg	B' row spread vs init
d=16 no G_t	315.46	314.09	285.33	—	—	—	—
d=16, m=64	348.59	350.14	315.33	1.0000	3.54	10.44	0.19× (collapsed)
d=24 no G_t	321.43	325.84	294.89	—	—	—	—
d=24, m=64	316.42	308.24	287.81	0.9995	3.10	22.39	9.68×

At d=16 the checkpoint shows the trap exactly: within-row structure large (std 0.904), across-row spread *fallen* from 0.0197 to 0.0037; the logit range over the 64 features under a one-hot q is 0.05, so Q_g is uniform to four decimals throughout. The gradient argument: with Q_g uniform the message is the row mean, so $\partial\,\mathrm{message}/\partial B'[k] = 1/m$, identical for every k; the only row-distinguishing term enters through the softmax Jacobian at the order of the row spread itself — and the shared L2 shrinks that spread further. Self-locking from any near-symmetric start.

Revival (both locks removed). Job 940918: B' init std 0.5, B' excluded from the L2, d=24, no damping: val **297.71**, train 292.26, test **272.82**, $H(Q_g)/\max = 0.9079$, glob msg 18.59 vs arc msg 18.49, ablation KL 2.391 (single seed). The head carries half the total message and improves d=24 by 7.4 % val / 7.5 % test. Verdict on §22.2 turns conditional: the head is not inherently dead weight — it works, but only when initialised and regularised *unlike the other factors of the graph*; under the uniform treatment the source itself applies, it is guaranteed to die.

Re-collapse under damping. In the 2×2 cells the same revived head partially re-collapses ($H(Q_g)/\max$ 0.9886 / 0.9741; seeds of the best cell: 0.7708, 0.9484) against 0.9079 undamped: damping slows $\bar{q}$, and the gradient through $\bar{q}$ is the only row-separating term — damping and the head compete for the same dynamics. On the damped record the head costs 7.7 % and doubles the seed variance.

5.6 Readout comparison record

PTB, d=256 base config: exact readout 1555.04 / 1508.89 after 2000 steps (794 s) vs MFVI 695.70 / 655.09 after 6000 steps (308 s), both TITAN RTX — per step the exact readout cost ∼7.8× here (397 vs 51 ms). The predecessor implementation (commit `57118de`) recorded 78× per step (1832 vs 23.6 ms) with its chunked exact readout; kept for the record, not directly comparable. Worked-example contrast (technical note §5, reproduced digit for digit): on the same slot the MFVI readout gives $\hat{p}(W_4) = (.460, .267, .007, .267)$ and the exact readout $(.365, .215, .206, .215)$ — the mixture-of-softmaxes readout is visibly flatter, consistent with escaping mean-field mode-seeking. No trained side-by-side in the working region exists yet (Experiment 3 proper remains open).

6 What is established vs. open

Table 2: Established claims.

claim	strongest evidence	replication
Pipeline sound; the failures were PT's	GPT 115.43 on identical loop/token set	certifying
Weight sharing is not the failure	Looped 126.46 at 309,600 non-emb	1 seed
Early PT lands on the unigram across the sweep	§4.1; ablation KL exactly 0.0000	15 configs, 1 seed each
Loss of cross-sequence content, no single broken stage	§4.2; untrained control carries 0.73 to logits	4 checkpoints
$\log \lvert D_t \rvert$ identity forces the unigram once $\bar{q}$ is position-flat	§4.1 identity; slots-vs-sequences std ratio 1700×	analytic + measured
Working region is joint in (d, lr)	§4.4 grids; oracle reached at d=16, lr 0.02	best cell, 3 seeds
Channels and width degrade independently	§4.5 cross grid at fixed lr	1 seed/cell
$\lVert G \rVert / \lVert S_w \rVert$ separates outcomes; within-run transition; two-sided	Figs. 1–2; every PTB run	∼30 runs
Freezing b helps; transition moves 2500 → 1000	473.28 → 430.04	**1 seed**
Damping α_Z=0.25 opens d=32; largest gain	641.5 → 250.69 ± 2.32	**3 seeds**
α_Z=0.5 insufficient	531.10; stall trace	1 seed
G_t dies by row collapse under uniform treatment	checkpoint spreads; gradient symmetry	2 configs, 1 seed
G_t revives when treated unlike other factors	297.71 at d=24; glob msg ≈ arc msg	**1 seed**
G_t on the damped record hurts and destabilises	+7.7 %; seed sd 5.36 vs 2.32	3 seeds (sd part)
G_t×floor interaction does *not* survive seeds	247.66 ± 5.36 vs 250.69 ± 2.32	3 seeds
Exact readout unusable as trained here at scale	1555.04 on PTB	1 seed

Table 3: Open questions, with measurement cost.

question	what is known	cost
The d=48 anomaly (non-monotone ladder at fixed α_Z)	344.31; only damped run with train > val	3 runs, ∼30 min
$\alpha^*(d)$ and the ceiling under damping	0.25 chosen at d=32	$d \times \alpha_Z$ grid, ∼2 h
Budget vs schedule (cosine caveat)	not separated	1 fixed-lr run
Does the frozen-b gain replicate?	single seed	2 runs
G_t revival replication; m-dependence	single seed; m=256 never run	4 runs
Matched-budget PT vs GPT/Looped	blocked on the two items above	depends
Beyond PTB; other tasks	nothing measured	hours
Experiment 3 proper (trained exact vs MFVI in the working region)	only §5.6 record	2 runs at exact cost
Dropout for PT (source: 0.15; placement unspecified)	now relevant: train < val	design + 2 runs

7 Corrections log

Claims this session itself made, later found wrong, and what replaced them — kept because it is what makes the rest of the record trustworthy.

8 Reproduction

Login-node venv (py3.11, torch 2.13 cpu) for tests; GPU jobs use the cluster anaconda (py3.8.8, torch 2.0.1+cu117). All batch scripts assert CUDA and the `/public` mount; exclude `ai_gpu32,ai_gpu33`. `python -m pytest`: 96 pass at `7cdcf3a`. Full argument dumps are stored verbatim in each result JSON under `"args"`; every run went through `sbatch ... <script>.slurm <args>`.

Comparability boundaries. (i) Generator bug, fixed at `0e679f3` (v0.8.2): `Dirichlet.sample` ignores its generator, so pre-fix synthetic-chain scripts drew different chains despite equal seeds (lr_probe oracle 3.84/6.52; region grid 1.96/3.09; deconfound, post-fix, 3.577/7.105). Across probes only gap-closed fractions are comparable, never raw perplexities; PTB numbers unaffected. (ii) JSONs written before `064db8d` store the trace *minimum* as val; from then on the final value, with the minimum kept alongside. Only `attack_final` d=32 is materially affected (523.08 stored vs 641.5 final; the trace carries the truth). (iii) The 940858 d=16 reference checkpoint was overwritten before the naming fix; its numbers come from logs/JSON, weight-level re-inspection needs re-training.

#	wrong claim	where it broke	replacement
1	Constraint 3 read as "no gradients into the prefix"	Part II §12.3 / §18 say the opposite	messages no, gradients yes; `detach_prefix=False` mainline
2	$\rho = 233$ (Lemma 23.1) as a stability finding; root column its cause	fixed-point probe: one fixed point at ρ=233; onset $\sim$130; root_init 1000$\times$ smaller changed nothing	vacuous bound; diagnostic only
3	Root column an "open defect" / attention sink	trained model avoids ROOT (0.15$\times$ uniform)	measured variable; logged
4	"Predecessor failed because MFVI rounds degrade fitting"	fit rate: γ=0 1/5 vs γ=3 4–5/5 at every round count	predecessor lacked RPE
5	Check 6 (overfit) on a single seed	predecessor's own trap repeated	multi-seed fit-rate test
6	"S does not grow" (from the ratio)	$\lVert S_w \rVert$ measured 17.4–20.1 vs init 0.32	withdrawn
7	Saturation as the cause	λ_Z sweep: entropy recovered, ppl unmoved, KL still 0	SUPERSEDED by the $\log \lvert D_t \rvert$ identity
8	Initialisation as the cause	context-using init still lands on the unigram	SUPERSEDED
9	"The failure is scale-dependent"	bisection ran at lr 10^{-3} only	joint (d, lr) region
10	$\lVert G \rVert / \lVert S_w \rVert$ band stated as two-sided (2–5)	best-yet run sat at 1.09 and was stopped as "out of band"	ceiling-only (≤ 5)
11	best-val = min over the trace	divergent d=32: min 523.08 at step 500, final 641.5	report final; keep min alongside
12	"G_t is dead weight"	revival at d=24: $H(Q_g)$ 0.908, $-7.4\,\%$	conditional verdict (§5.5)
13	Six job deaths called "transient scheduler faults"	all six on `ai_gpu32/33`: `/public` unmounted; `cd` fails under `set -e`; no output file possible	mount check in all batch scripts
14	"Paired sbatch submissions fail"	single submissions failed identically there	node fault
15	Checkpoint name without the run tag	G_t run overwrote its d=16 reference	tag in name; that checkpoint lost (numbers from logs/JSON)
16	"New record 241.89; the interaction is real"	seeds: 247.66 ± 5.36 vs 250.69 ± 2.32	withdrawn
17	d=48/64 non-emb params 24,672 / 32,896 (memory)	slurm dumps: 36,960 / 65,664	corrected everywhere

A Experiment 0 validation record

Checks 1–9 of the research plan pass (96 behavioural tests); the worked example of the technical note §5 reproduces digit for digit (filtering marginals $(.818, .182)/(.957, .043)/(.006, .994)$; attention logits $(.160, .413, .178, 1.776)$ and $(.074, .387, .128, 1.890)$; $\hat{p}(W_4) = (.460, .267, .007, .267)$). Exact readout vs brute-force enumeration of the slot joint agrees to 10^{-12} in float64, including with the global head folded in as an extra leaf; the `logcumsumexp` scan is checked against slot-by-slot assembly for $\gamma \in \{0, 1, 2, 4, 9\}$. The free-energy check carries a mutation test: a sign-flipped H-update fails it while checks 1–7 pass unchanged.

Fixed-point multiplicity vs Lemma 23.1's ρ (48 random starts, float64): $\rho = 0.05 \to 1$ fixed point; $103 \to 1$; $135 \to 2$; $293 \to$ up to 5; $10411 \to$ 15–22 (separation 1.0). Source-config row (d=384, h=16, r=64, untrained): $\rho = 233 \to$ *one* fixed point; shrinking root_init_std 1000$\times$ at init 2.0 leaves $\rho \approx 9500$ and the count unchanged — $\rho \geq 1$ is vacuous here, not an instability finding. Untrained prefix ablation: KL $9.5 \cdot 10^{-11}$ at init 0.02 vs $6.9 \cdot 10^{-2}$ at 0.5 (argmax moves on 62 % of slots). Schedule divergence (parallel vs serial, d=32): TV $3.4 \cdot 10^{-6}$ at init 0.02, 0.682 with $\mathrm{TV}_{\max} = 1.000$ at 2.0 (the multistable regime). ROOT column at init sits 121$\times$ above the contracted prefix rows; after training the model *avoids* ROOT (0.15$\times$ uniform).

Table 4: Result → job → commit (commit = where the result JSON was added).

result (JSON tag)	job	HW	commit
pilot_mfvi_T3 / pilot_exact_T3	940422 / 940423	TITAN	f492a9d / 0296af9
probe_T1 / probe_l2_5 / probe_lambdaZ4	940436 / 940435 / 940438	T/T/A40	0296af9
sweep_lz16/32/64	940483/84/85	T/T/A40	0231ce7
step1_nob / step1_withb	940444 / 940445	TITAN	e62a233
init0.2 / init0.5	940489 / 940490	TITAN	0231ce7
tau1 / tau4 / tau8	940502/503/504	T/T/A40	e24a0d4
base_gpt / base_looped / base_pt / scale	940799	A40	8842e2c
lr_probe	940809	TITAN	0cc04f5
region_probe (grid, replicate, PTB)	940814	A40	35f10cd
deconfound_dh	940827	A40	bee3b36
attack (3a/3b)	940844	A40	779a0e4
attack_freezeb / attack_long	940848 / 940850	A40	b2f6508
attack_final (ladder 16/24/32)	940858	A40	064db8d
G_R1 / G_R2	940872 / 940879	A40	1ecd1fa
damp d32 a0.5/a0.25, d48, d64	940913/914/915/916	TITAN	64ef791
G_rev_d24	940918	TITAN	b49933d
combo / full / floor (2×2)	940920 / 940921 / 940927	TITAN	b49933d
RECseed1/2, Bseed1/2	940932/934, 940931/933	TITAN	7cdcf3a

Table 5: Fit rate vs RPE and round count (5 seeds, 1200 Adam steps, d=16, h=2, V=12; the predecessor implementation had no RPE).

	T=1	T=2	T=3	T=4	T=5
$\gamma = 0$ (no RPE), fit rate	1/5	2/5	1/5	1/5	1/5
$\gamma = 0$, median final loss	0.276	0.146	0.175	0.384	0.175
$\gamma = 3$, fit rate	4/5	5/5	4/5	4/5	4/5
$\gamma = 3$, median final loss	0.000	0.000	0.000	0.000	0.000

Figures

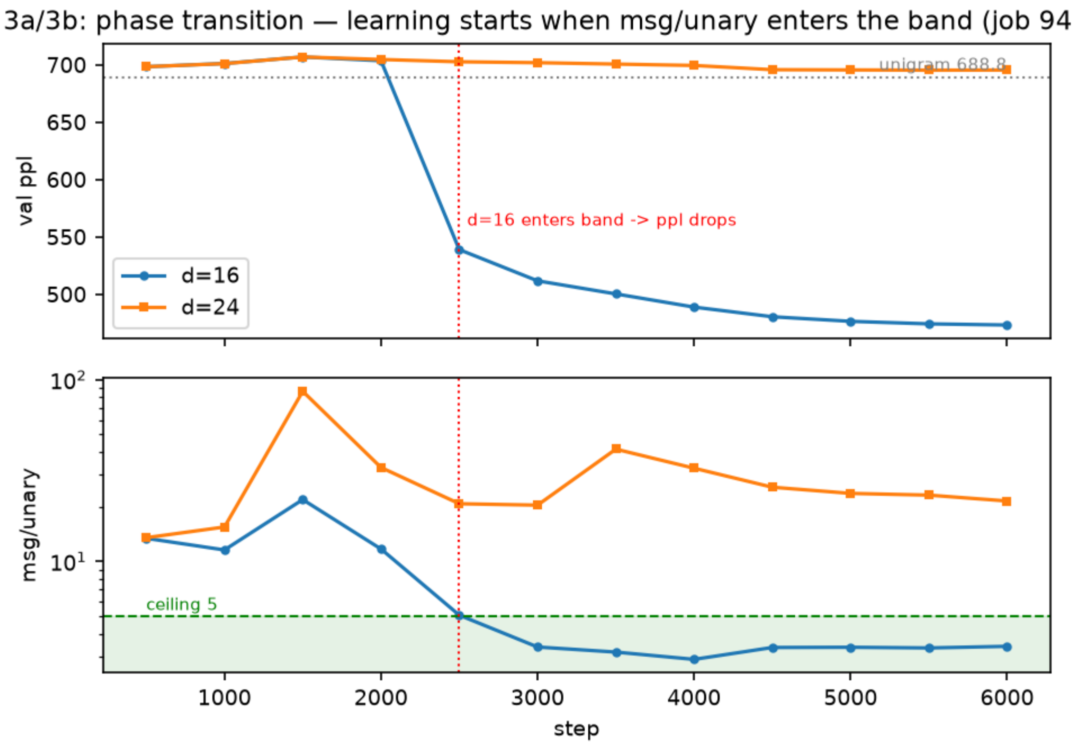

Figure 1: The phase transition (job 940844): validation perplexity and $\|G\|/\|S_w\|$ on one time axis, d=16 vs d=24. Learning begins at the evaluation where $\|G\|/\|S_w\|$ first enters the band; at d=24 it never enters and perplexity never leaves the unigram.

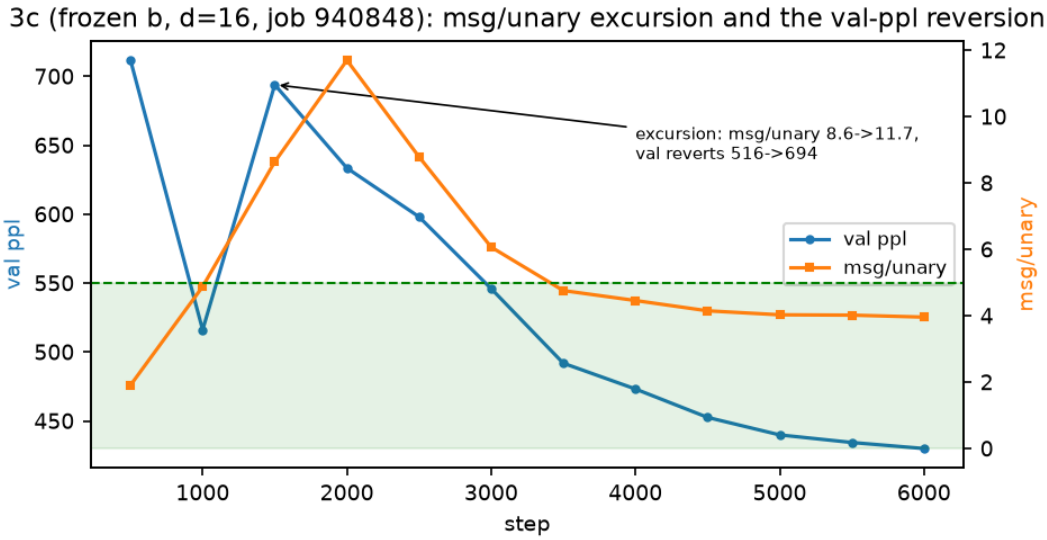

Figure 2: The 3c excursion (job 940848, frozen b): $\|G\|/\|S_w\|$ leaves the band at steps 1500–2000 and validation perplexity reverts 516 $\rightarrow$ 694, then falls monotonically once the scalar re-enters — the two-sided evidence.

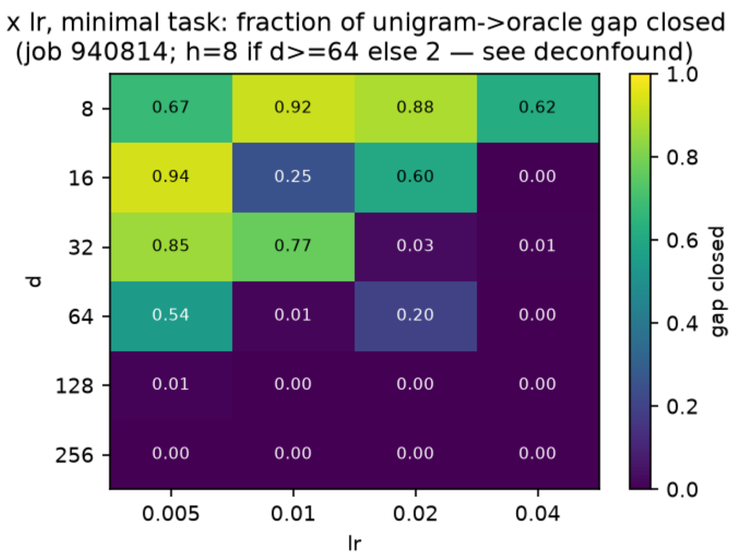

Figure 3: $d \times \mathrm{lr}$ on the minimal task (job 940814): fraction of the unigram-to-oracle gap closed.

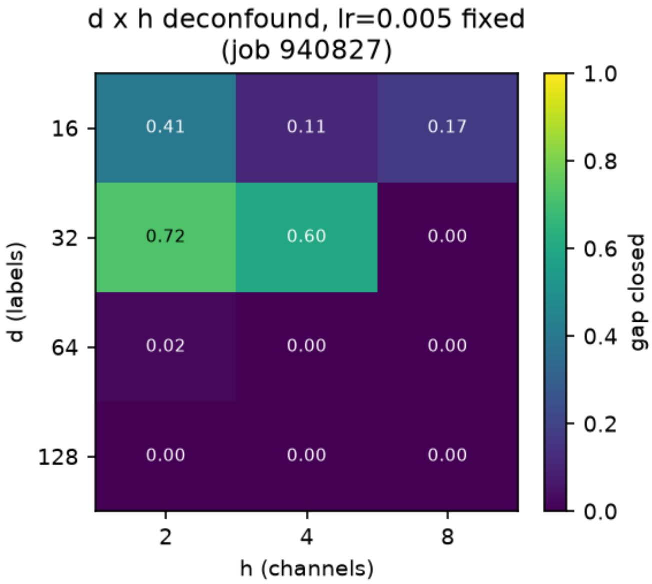

Figure 4: $d \times h$ deconfound at fixed lr (job 940827): channels and width degrade the model independently.

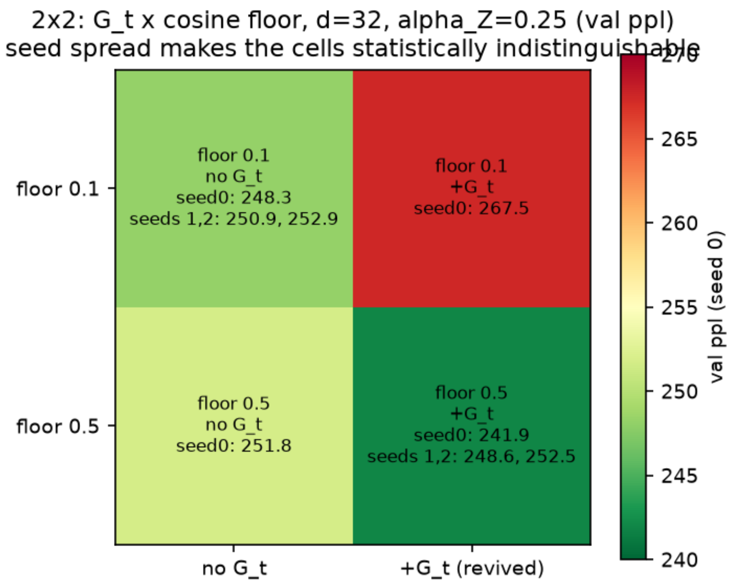

Figure 5: The $G_t\times$ floor square at d=32, α_Z=0.25, with seed replication: the apparent interaction does not survive the seed spread.

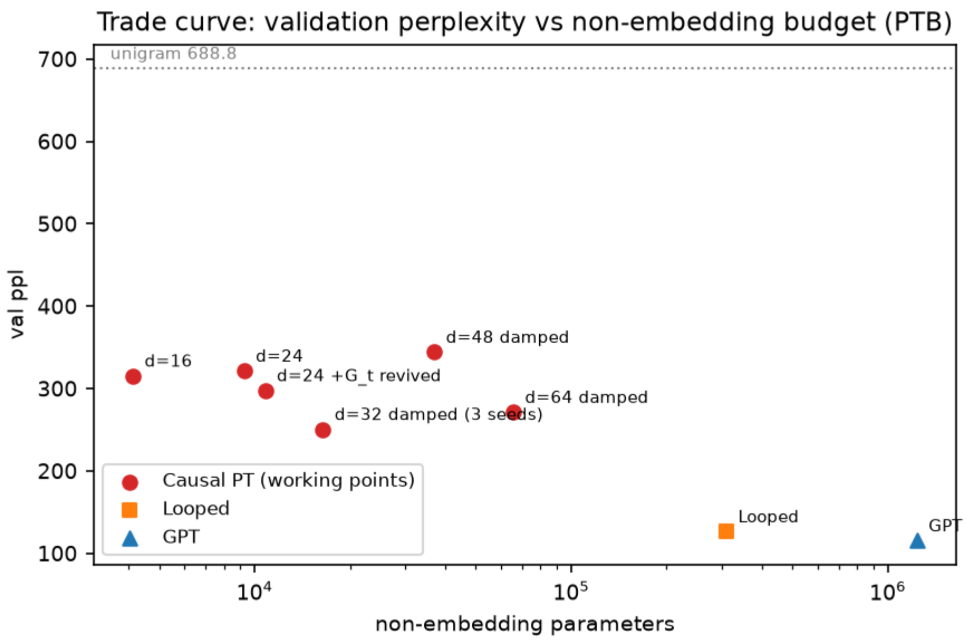

Figure 6: Trade curve: validation perplexity vs non-embedding parameters. PT working points (record with 3-seed error bar), GPT, Looped, unigram line.